

DATABASE DESIGN

Online job portal, the database design should include entities that capture key aspects like users, jobs, applications, and companies. Below is a suggested database schema:

1. Users Table

Stores information about users (job seekers and recruiters).

Column	Data Type	Description
user_id	INT (PK)	Unique identifier for each user
first_name	VARCHAR(50)	First name of the user
last_name	VARCHAR(50)	Last name of the user
email	VARCHAR(100)	User's email address (unique)
password	VARCHAR(100)	password Encrypted
phone	VARCHAR(20)	Contact number
user_type	ENUM('job_seeker','recruiter')	Type of user (Job Seeker or Recruiter)
created_at	TIMESTAMP	Account creation timestamp
updated_at	TIMESTAMP	Account creation timestamp

2. UserProfile Table

Stores additional information about job seekers.

Column	Data Type	Description
Profile_id	INT(PK)	Unique identifier for the profile
User_id	INT (FK)	Reference to user_id in Users
resume	VARCHAR(255)	File path to the resume
skills	TEXT	Skills of the user
experience	TEXT	Work experience details
education	TEXT	Education background
location	VARCHAR(100)	User's current location

3. Companies Table

Stores information about companies that are hiring.

Column	Data Type	Description
company_id	INT (PK)	Unique identifier for each company
user_id	INT (FK)	Reference to user_id in Users (Recruiter)
name	VARCHAR(100)	Name of the company
description	TEXT	Description of the company
location	VARCHAR(100)	Location of the company
website	VARCHAR(100)	Website URL of the company

4. Jobs Table

Stores information about job postings.

Column	Data Type	Description
job_id	INT (PK)	Unique identifier for each job
company_id	INT (FK)	Reference to company_id in Companies
title	VARCHAR(100)	Reference to company_id in Companies
description	TEXT	Job description
requirements	TEXT	Required skills and qualifications
location	VARCHAR(100)	Job location
salary	VARCHAR(50)	Salary range
employment_type	ENUM('full-time', 'part-time', 'contract', 'internship')	Type of employment
created_at	TIMESTAMP	Job posting creation timestamp
updated_at	TIMESTAMP	Last update timestamp

5. Applications Table

Tracks job applications submitted by job seekers.

Column	Data Type	Description
application_id	INT (PK)	Unique identifier for each application
job_id	INT (FK)	Reference to job_id in Jobs
user_id	INT (FK)	Reference to user_id in Users (Job Seeker)
Status	ENUM('applied', 'interview', 'hired', 'rejected')	Status of the application
applied_at	TIMESTAMP	Application submission timestamp

6. JobCategories Table

Stores different categories or sectors for jobs.

Column	Data Type	Description
category_id	INT (PK)	Unique identifier for each category
name	VARCHAR(100)	Name of the category (e.g., IT, Marketing)

7. JobCategoryMapping Table

Associates jobs with one or more categories.

Column	Data Type	Description
job_id	INT (FK)	Reference to job_id in Jobs
category_id	INT (FK)	Reference to category_id in JobCategories

Relationships:

- **Users:** Can be either job seekers or recruiters.
- **Companies:** Linked to recruiters through user_id.
- **Jobs:** Linked to companies and can belong to multiple categories.
- **Applications:** Linked to job seekers and specific jobs.

This design provides a solid foundation for implementing the main functionalities of an online job portal, like user registration, job posting, searching jobs, and applying for them.